

HRISHI BHATTACHARYYA

Senior Full Stack Developer | Engineering Team Lead

Kolkata, West Bengal, India | +91 6294660141 | hrishisgp97@gmail.com
[linkedin.com/in/hrishi-bhattacharyya-b78332204](https://www.linkedin.com/in/hrishi-bhattacharyya-b78332204) | github.com/Hrishi1176

PROFESSIONAL SUMMARY

Senior Full Stack Developer and Engineering Team Lead with 4+ years of experience designing scalable web architectures, leading cross-functional development teams, and integrating AI/LLM capabilities into production systems. Proficient across the full stack with React.js, Next.js, Node.js, and Python, with hands-on expertise in RAG pipelines, MCP server integration, prompt engineering, ETL data pipelines, and Apache Superset BI dashboards. Track record of translating business requirements into scalable architecture, managing multi-tenant backend systems, and shipping end-to-end products from database design through cloud deployment.

TECHNICAL SKILLS

Languages & Frontend: JavaScript (ES6+), TypeScript, React.js, Next.js 16, HTML5, CSS3, Tailwind CSS

Backend & APIs: Python, Node.js, Express.js, FastAPI, AWS Chalice, REST APIs, WebSockets, Socket.IO

AI & LLM Engineering: LLM Integration, RAG (Retrieval-Augmented Generation), MCP (Model Context Protocol), Prompt Engineering, Groq Llama 3, Google MediaPipe

Databases: MongoDB, MySQL, PostgreSQL, Snowflake

Cloud & DevOps: AWS EC2, DigitalOcean, Docker, Nginx, CI/CD Pipelines

Tools & Practices: Git/GitHub, System Design, Agile/DevOps, ETL Pipelines, Apache Superset, JWT & NextAuth.js Authentication

PROFESSIONAL EXPERIENCE

Manager & Senior Software Developer

eDge Wrapper Technology Pvt. Ltd. | 2023 – June 2026

- Led and managed cross-functional development teams — assigning tasks, conducting code reviews, and ensuring on-time delivery across multiple concurrent client projects.
- Designed scalable application architectures and technical structures aligned to business and client requirements, acting as lead technical decision-maker for project structure.
- Designed data models and built reporting solutions to support business intelligence and data-driven decision-making.
- Implemented AI/LLM integrations using Python-based backend services, including Retrieval-Augmented Generation (RAG) pipelines for efficient, contextual AI responses.
- Integrated MCP (Model Context Protocol) servers, enabling AI systems to access and retrieve data efficiently across multiple data sources.
- Developed dynamic, optimized prompts that improved AI output accuracy while reducing token usage and cost.
- Built scalable backend services for booking management and delivery-API integrations for e-commerce clients within a sandboxed multi-tenant environment.
- Built and maintained ETL pipelines for data processing, transformation, and integration, feeding Apache Superset dashboards used for business reporting.

Associate Software Developer

eDge Wrapper Technology Pvt. Ltd. | 2022 – 2023

- Developed full-stack web applications using React.js and Next.js on the frontend and Node.js, Python, AWS Chalice, and FastAPI on the backend.
- Integrated and maintained multiple database systems — MongoDB, PostgreSQL, and Snowflake — across client projects.
- Deployed and maintained applications using CI/CD pipelines, Docker, Nginx, DigitalOcean Droplets, and AWS EC2.
- Performed application testing using various frameworks based on client and project requirements.
- Built and maintained WordPress-based e-commerce websites, and supported Magento version upgrades, maintenance, and bug fixes.

KEY PROJECTS

ZodiacSense — AI-Powered Astrology & Palmistry Platform

Tech: Next.js 16, React 19, TypeScript, MongoDB, NextAuth.js, MediaPipe, Framer Motion

- Architected a full-stack astrology platform delivering personalized horoscopes, precise birth-chart generation using Swiss Ephemeris data, and interactive geolocation-based birth-location mapping.
- Built a real-time AI palm-reading feature using Google's MediaPipe hand-tracking framework to analyze palm lines from a live device camera feed.
- Developed an interactive AI astrology chatbot and a marriage-compatibility engine that compares birth charts using traditional astrological principles.
- Implemented full internationalization with real-time translation, NextAuth.js authentication, and on-demand exportable PDF astrology reports.

OpenConnect — Secure Remote Desktop Support Application

Tech: Electron, Express.js, Socket.IO, Python, WebRTC

- Built an open-source remote support tool requiring explicit host approval before any session activates, with pending viewer requests auto-expiring after 30 seconds.
- Implemented WebRTC-first peer-to-peer streaming with automatic WebSocket relay fallback for restrictive network environments.
- Hardened the Electron client with contextIsolation, sandboxing, disabled Node integration, and blocked external navigation to reduce attack surface.
- Secured the Socket.IO signaling server with helmet headers, CSP, HSTS, rate limiting, and strict CORS; added server-side validation for all control, chat, and file-transfer payloads with a 5MB transfer cap.

Vibely — AI-Integrated Social Media Platform (PWA)

Tech: Python, FastAPI, React 19, PostgreSQL, Groq Llama 3 API

- Built an AI-powered social platform on a FastAPI backend and React 19 frontend, engineered to run entirely on free-tier cloud infrastructure.
- Integrated the Groq Llama 3 API to power AI caption generation, vibe-tag recommendations, automated content moderation, and an interactive AI chat companion.
- Implemented JWT authentication with bcrypt hashing and enforced daily usage quotas (10 AI generations, 15 posts, 10 image uploads per user) plus 60 requests/minute rate limiting to keep infrastructure costs near \$0.
- Shipped as an installable Progressive Web App with a custom dark-glassmorphism design system.

NexusWork — Co-Working Space Desk & Room Booking Platform

Tech: Node.js, Express, MongoDB, React, Vite, Tailwind CSS, Docker

- Designed a full-stack booking system with role-based access control across Visitor, Member, and Admin roles, including a real-time hourly availability calendar.
- Implemented a 2-layer authentication model using httpOnly refresh-token cookies plus in-memory access tokens, eliminating localStorage-based token exposure.
- Engineered a concurrency-safe reservation engine with mutex locks and database-level transactions; validated correctness with a stress test firing 10 simultaneous booking requests, confirming exactly 1 success and 9 correctly rejected conflicts.
- Structured the codebase with an OOP service/repository pattern on both backend and frontend, and delivered Swagger/OpenAPI docs, a Postman collection, and a full automated test suite (16/16 passing).
- Enabled one-command deployment via Docker Compose across MongoDB, Express, and Nginx-served React services.

EDUCATION

Bachelor of Engineering (B.E.), Information Technology

University Institute of Technology, The University of Burdwan | 2018 – 2021

CGPA: 7.5 / 10 (Grade A+)

Diploma in Engineering, Computer Science & Technology

Siliguri Government Polytechnic | Aug 2015 – Aug 2018

CGPA: 6.6 / 10 (Grade A)